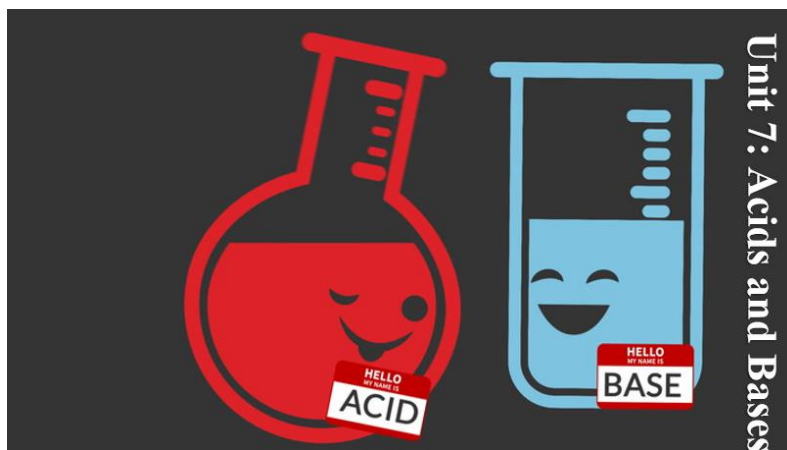


Name: _____ Period: _____



Unit 7: Acids and Bases

AP Classroom Topic 8

Unit Map

Textbook Reference: Zumdahl *Chemistry* Chapter 15-16

Date	Day	Learning Targets	Activities	Work on this
1/23-26	U7D1	What are Arrhenius and Bronsted-Lowry acids and bases?	Exam 6 Begin listening to lectures #1&2	Listen to lectures 1 & 2 for quiz next time
1/27-28	U7D2	How is pH calculated?	Open Note Quiz Inquiry Investigation: temperature and pH	Textbook Problem Set #1
1/29-30	U7D3	What is the pH of a weak acid?	Inquiry Investigation: percent dissociation	Textbook Problem Set #1
2/2-3	U7D4	What is the pH of a weak base?		Textbook Problem Set #1 Quiz next time!
2/4-5	U7D5	What is an acidic or basic salt?	Quiz!	Textbook Problem Set #2
2/6-9	U7D6	What is a buffer?	Titration Simulation	Textbook Problem Set #2
2/10-11	U7D7	What is a titration curve?	Titration curve analysis	Textbook Problem Set #2
2/12-13	U7D8	Review		Textbook Problem Set #2
2/18-19	U8D1	What is atomic structure?		Textbook Problem Set #2
2/20-23	U7D9	Acid Base Exam		

Unit 7: Acids and Bases Learning Targets (CED Topic 8, partial 4)

4.8	Introduction to Acid Base Reactions
4.8A	Identify species as Brønsted-Lowry acids, bases, and / or conjugate acid-base pairs, based on proton-transfer involving those species.
	4.8.A.1: By definition, a Brønsted-Lowry acid is a proton donor and a Brønsted-Lowry base is a proton acceptor.
	4.8.A.2: Only in aqueous solutions, water plays an important role in many acid-base reactions, as its molecular structure allows it to accept protons from and donate protons to dissolved species.
	4.8.A.3: When an acid or base ionizes in water, the conjugate acid-base pairs can be identified and their relative strengths compared. ★
8.1	Calculate the values of pH and pOH, based on K_w and the concentration of all species present in a neutral solution of water.
	8.1.A.1: The concentrations of hydronium ion and hydroxide ion are often reported as pH and pOH, respectively.
	The terms “hydrogen ion” and “hydronium ion” and the symbols H^+ (aq) and H_3O^+ (aq) are often used interchangeably for the aqueous ions of hydrogen. Hydronium ion and H_3O^+ (aq) are preferred, but H^+ (aq) is also accepted on the AP Exam.
	8.1.A.2: Water autoionizes with an equilibrium constant K_w .
	8.1.A.3: In pure water, $pH = pOH$ is called a neutral solution. At $25^\circ C$, $pK_w = 14.0$ and thus $pH = pOH = 7.0$.
8.2	pH and pOH of Strong Acids and Bases
8.2A	Calculate the pH and pOH based on concentrations of all species in a solution of a strong acid or a strong base.
	8.2.A.1: Molecules of a strong acid (e.g., HCl, HBr, HI, $HClO_4$, H_2SO_4 , and HNO_3) will completely ionize in aqueous solution to produce hydronium ions and the conjugate base of the acid. As such, the concentration of H_3O^+ in a strong acid solution is equal to the initial concentration of the strong acid, and thus the pH of the strong acid solution is easily calculated.
	8.2.A.2: When dissolved in solution, strong bases (e.g., group I and II hydroxides) completely dissociate to produce hydroxide ions. As such, the concentration of OH^- in a strong base solution is equal to the initial concentration of a group I hydroxide and double the initial concentration of a group II hydroxide, and thus the pOH (and pH) of the strong base solution is easily calculated.
8.3	Weak Acid and Base Equilibria
8.3A	Explain the relationship among pH, pOH, and concentrations of all species in a solution of a monoprotic weak acid or weak base.
	8.3.A.1: Weak acids react with water to produce hydronium ions. However, only a small percentage of molecules of a weak acid will ionize in this way. Thus, the concentration of H_3O^+ is much less than the initial concentration of the molecular acid, and the vast majority of the acid molecules remain un-ionized.
	8.3.A.2: A solution of weak acid involves equilibrium between an un-ionized acid and its conjugate base. The equilibrium constant for this reaction is K_a , often reported as pK_a . The pH of a weak acid solution can be determined from the initial acid concentration and the pK_a .
	8.3.A.5: The percent ionization of a weak acid (or base) can be calculated from its pK_a (pK_b) and the initial concentration of the acid (base). The percent ionization can also be calculated from the initial concentration of the acid (base) and the equilibrium concentration of any of the species in the equilibrium expression.
	8.3.A.6: For any conjugate acid-base pair, the acid ionization constant and base ionization constant are related by K_w :
8.4	Acid and Base Reactions and Buffers
8.4A	Explain the relationship among the concentrations of major species in a mixture of weak and strong acids and bases.
	8.4.A.1: When a strong acid and a strong base are mixed, they react quantitatively in a reaction represented by the equation: H^+ (aq) + OH^- (aq) $\rightarrow$ H_2O (l).
	The pH of the resulting solution may be determined from the concentration of excess reagent.
	8.4.A.2: When a weak acid and a strong base are mixed, they react quantitatively in a reaction represented by the equation: HA (aq) + OH^- (aq) $\rightleftharpoons$ A^- (aq) + H_2O (l).
	If the weak acid is in excess, then a buffer solution is formed, and the pH can be determined from the Henderson-Hasselbalch (H-H) equation (see 8.9.A.1). If the strong base is in excess, then the pH can be determined from the moles of excess hydroxide ion and the total volume of solution. If they are equimolar, then the (slightly basic) pH can be determined from the equilibrium represented by the equation: A^- (aq) + H_2O (l) $\rightleftharpoons$ HA (aq) + OH^- (aq).
	8.4.A.3: When a weak base and a strong acid are mixed, they will react quantitatively in a reaction represented by the equation: B (aq) + H_3O^+ (aq) $\rightleftharpoons$ HB^+ (aq) + H_2O (l).
	If the weak base is in excess, then a buffer solution is formed, and the pH can be determined from the H-H equation. If the strong acid is in excess, then the pH can be determined from the moles of excess hydronium ion and the total volume of solution. If they are equimolar, then the (slightly acidic) pH can be determined from the equilibrium represented by the equation: HB^+ (aq) + H_2O (l) $\rightleftharpoons$ B (aq) + H_3O^+ (aq).
	8.4.A.4: When a weak acid and a weak base are mixed, they will react to an equilibrium state whose reaction may be represented by the equation: HA (aq) + B (aq) $\rightleftharpoons$ A^- (aq) + HB^+ (aq).

8.5	Acid Base Titrations
8.5A	Explain results from the titration of a mono- or polyprotic acid or base solution, in relation to the properties of the solution and its components.
	8.5.A.1: An acid-base reaction can be carried out under controlled conditions in a titration. A titration curve, plotting pH against the volume of titrant added, is useful for summarizing results from a titration.
	8.5.A.2: At the equivalence point for titrations of monoprotic acids or bases, the number of moles of titrant added is equal to the number of moles of analyte originally present. This relationship can be used to obtain the concentration of the analyte. This is the case for titrations of strong acids / bases and weak acids / bases.
	8.5.A.3: For titrations of weak acids / bases, it is useful to consider the point halfway to the equivalence point, that is, the half-equivalence point. At this point, there are equal concentrations of each species in the conjugate acid-base pair, for example, for a weak acid $[HA] = [A^-]$. Because $pH = pK_a$ when the conjugate acid and base have equal concentrations, the pK_a can be determined from the pH at the half-equivalence point in a titration.
	8.5.A.4: At the equivalence point, pH is determined by the major species in solution. Strong acid and strong base titrations result in neutral pH at the equivalence point. However, in titrations of weak acids (weak bases), the conjugate base of the weak acid (conjugate acid of the weak base) is present at the equivalence point and can undergo proton-transfer reactions with the surrounding water, producing basic (acidic) solutions.
	8.5.A.5: For polyprotic acids, titration curves can be used to determine the number of acidic protons. In doing so, the major species present at any point along the curve can be identified, along with the pK_a associated with each proton in a weak polyprotic acid. ★
8.6	Molecular Structures of Acids and Bases
8.6A	Explain the relationship between the strength of an acid or base and the structure of the molecule or ion.
	8.6.A.1: The protons on a molecule that will participate in acid-base reactions, and the relative strength of these protons, can be inferred from the molecular structure.
	Strong acids (such as HCl, HBr, HI, HClO ₄ , H ₂ SO ₄ , and HNO ₃) have very weak conjugate bases that are stabilized by electronegativity, inductive effects, resonance, or some combination thereof.
8.7	pH and pK_a
8.7A	Explain the relationship between the predominant form of a weak acid or base in solution at a given pH and the pK_a of the conjugate acid or the pK_b of the conjugate base.
	8.7.A.1: The protonation state of an acid or base (i.e., the relative concentrations of HA and A ⁻) can be predicted by comparing the pH of a solution to the pK_a of the acid in that solution. When solution $pH < acid\ pK_a$, the acid form has a higher concentration than the base form. When solution $pH > acid\ pK_a$, the base form has a higher concentration than the acid form.
	8.7.A.2: Acid-base indicators are substances that exhibit different properties (such as color) in their protonated versus deprotonated state, making that property respond to the pH of a solution.
	8.7.A.3: To ensure accurate results in a titration experiment, acid-base indicators should be selected that have a pK_a close to the pH at the equivalence point.
8.8	Properties of Buffers
8.8A	Explain the relationship between the ability of a buffer to stabilize pH and the reactions that occur when an acid or base is added to a buffered solution.
	8.8.A.1: A buffer solution contains a large concentration of both members in a conjugate acid-base pair. The conjugate acid reacts with added base and the conjugate base reacts with added acid. These reactions are responsible for the ability of a buffer to stabilize pH.
8.9	Henderson -Hasselbach Equation
8.9A	Identify the pH of a buffer solution based on the identity and concentrations of the conjugate acid-base pair used to create the buffer.
	8.9.A.1: The pH of the buffer is related to the pK_a of the acid and the concentration ratio of the conjugate acid-base pair. This relation is a consequence of the equilibrium expression associated with the dissociation of a weak acid, and is described by the Henderson-Hasselbalch equation. Adding small amounts of acid or base to a buffered solution does not significantly change the ratio of $[A^-] / [HA]$ and thus does not significantly change the solution pH. The change in pH on addition of acid or base to a buffered solution is therefore much less than it would have been in the absence of the buffer. ★ ★
8.10	Buffer Capacity
8.10A	Explain the relationship between the buffer capacity of a solution and the relative concentrations of the conjugate acid and conjugate base components of the solution.
	8.10.A.1: Increasing the concentration of the buffer components (while keeping the ratio of these concentrations constant) keeps the pH of the buffer the same but increases the capacity of the buffer to neutralize added acid or base.
	8.10.A.2: When a buffer has more conjugate acid than base, it has a greater buffer capacity for addition of added base than acid. When a buffer has more conjugate base than acid, it has a greater buffer capacity for addition of added acid than base.



Name it!

- a) HBr
- b) H_2CO_3
- c) HClO
- d) H_2S

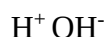
Review: Naming Acids

Formula!

- a) Phosphoric acid
- b) Sulfurous acid
- c) Hydroiodic acid
- d) Acetic Acid

Arrhenius Acids and Bases

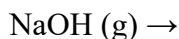
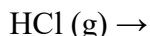
Recall that water, H_2O , can be thought of as



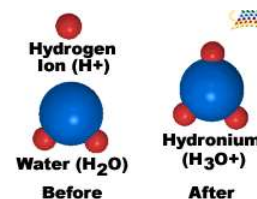
Define an acid and a base, according to the Arrhenius definition.

- acid =
- base =
-

Write the products of each reaction below, which involves the dissociation of each reactant into aqueous ions.



hydronium ion (_____)- formed on reaction of a proton with a water molecule. _____ and _____ are used interchangeably in most situations.



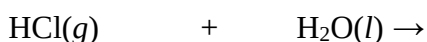
Brønsted-Lowry Acids and Bases

Brønsted-Lowry Acids and Bases – uses a _____ of acid and base to understand the _____ of a proton, H^+ , from the _____ to the _____.

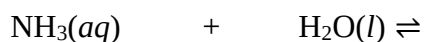
Brønsted-Lowry Acid – particle that _____ to another particle (the base)

Brønsted-Lowry Base – particle that _____ from another particle (the acid)

Write the formulas for the two products formed in these chemical reaction.



HCl behaves as an _____ because it _____ H^+ , and water behaves as a _____ because it _____ H^+ .



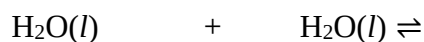
NH_3 behaves as a _____ because it _____ H^+ , and water behaves as a _____ because it _____ H^+ .

A Brønsted-Lowry acid must have an _____ atom that it can lose as a positive ion (sometimes called the labile proton)

A Brønsted-Lowry base must have a _____ that it can use to bind the H^+ ion.

Water is an example of an amphiprotic substance. What does it mean to be amphiprotic?

Write an equation showing water acting as an amphoteric substance:



Conjugate acids and bases



The products of a B-L acid base reaction are often referred to as the _____ of the reaction.

conjugate base- everything that remains of the acid molecule after a _____

conjugate acid- base plus _____

Practice:

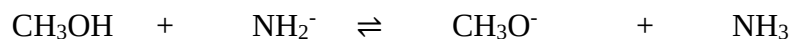
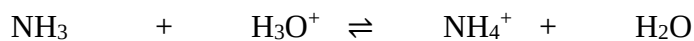
Identify the conjugate bases of these substances.

- HCl
- HNO_3
- NH_3
- $\text{HC}_2\text{H}_3\text{O}_2$
- CH_3NH_3^+

Identify the conjugate acids of these substances.

- F^-
- NH_3
- H_2O

Practice: For the following reactions, identify the acid, base, conjugate acid and conjugate base:



Strong and Weak Acids and Bases

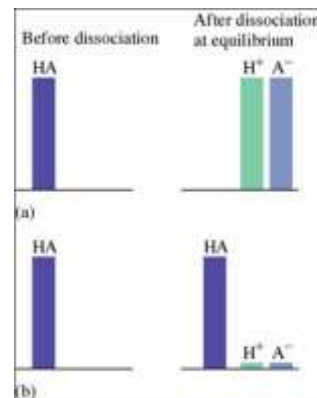
Strong acid – _____ K value, mostly dissociated substance.

Write a dissociation reaction for the strong acid, HX:

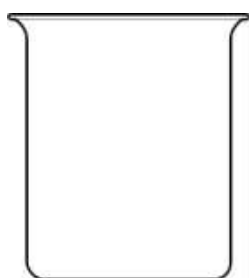
- equilibrium lies far to the right
- a strong acid yields a _____ conjugate base (much weaker than H_2O)
- Notice the single arrow! $K > 10^5$

Weak acid- _____ K value for dissociation, mostly undissociated. Write dissociation reaction for the weak acid, HA:

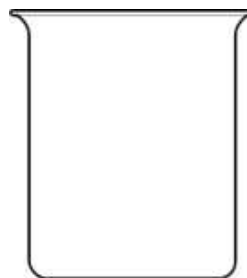
- equilibrium lies far to the left
- has a ' _____ ' conjugate base (stronger than water, but not a “strong” base)
- (strong) would look like in a beaker of water.
- Notice the single arrow! $K < 10^{-5}$



Sketch what five molecules of HF (weak) and five molecules of HCl



HF



HCl

8.2 pH and OH of Strong Acids and Bases

Reminder: Common strong acids -all completely dissociate! (Know these!)

Write the conjugate bases for each:

HCl (hydrochloric)

HNO_3 (nitric)

HClO_4 (perchloric)

HI (hydroiodic)

HBr (hydrobromic)

H_2SO_4 (sulfuric)

Sulfuric acid is a _____ acid which means that it has _____ acidic protons. The first (H_2SO_4) is strong and the second (HSO_4^-) is weak.

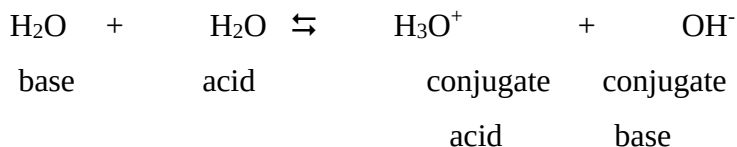
Write equations for the first and second dissociations of sulfuric acid in water.

Strong Bases Completely Dissociate

The most common strong bases are the hydroxide compounds of the alkali metals, such as _____ and _____. Other strong bases include hydroxide compounds of the heavier alkaline earth metals, such as _____, _____, and _____.

Write the dissociation for $\text{Ca}(\text{OH})_2$. What do you notice about the $[\text{OH}^-]$ for the alkaline earth metal hydroxides?

Autoionization of Water



Write the equilibrium expression for the dissociation of water, K_w :

At 25°C , $K_w = 1 \times 10^{-14}$ because $[\text{H}^+] = [\text{OH}^-] = 1 \times 10^{-7} \text{ M}$ (This value is on your pink sheet!)

This is a very important relationship that allows you to calculate $[\text{H}^+]$, $[\text{OH}^-]$, pH, and pOH.

No matter what an aqueous solution contains, at 25°C ,

$$[\text{H}^+][\text{OH}^-] = 1 \times 10^{-14}$$

Neutral solution $[\text{H}^+] = [\text{OH}^-]$ and pH ____ 7

Acidic solution $[\text{H}^+] > [\text{OH}^-]$ and pH ____ 7

Basic solution $[\text{H}^+] < [\text{OH}^-]$ and pH ____ 7

K_w varies with temperature-this can affect the pH of pure water at different temperatures!

The pH Scale

_____ is defined as the negative logarithm (in base 10) of the concentration of H^+ (or H_3O^+).

$$\text{pH} = -\log[\text{H}^+] \quad \text{or} \quad \text{pH} = -\log[\text{H}_3\text{O}^+]$$

For example, the pH of a solution in which $[\text{H}^+] = 1.0 \times 10^{-7} = 7.00$

Mathematically, this also means that concentration of H^+ can be found from pH:

$$[\text{H}^+] = \underline{\hspace{2cm}}$$

Significant figures in pH and other log values:

The number of decimal places in the log value should equal the number of significant digits in the original number (concentration).

Practice: pH = 4.42 How many significant figures??

Practice: Calculate the pH of a solution where $[H^+] = 1.3 \times 10^{-5}M$

pH and pOH

In chemistry, we use the convention that the “negative log” of some quantity is labeled p(quantity).

Therefore the concentration of hydroxide ions in solution is expressed in terms of the_____.

Similarly, $pK_w = -\log(K_w) = ______$ at 25°C and You should also know that $pH + pOH = ______$ at 25°C

- pH and pOH are logarithmic functions. The pH changes by 1 for every power of 10 change in $[H^+]$. pH decreases as $[H^+]$ increases.
- A substance with a pH of 3.00 has ten times the $[H^+]$ as a substance with a pH of 4.00.

From your formula sheet:

$$K_w = [H^+][OH^-] = 1.0 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

$$= K_a \times K_b$$

$$pH = -\log[H^+], pOH = -\log[OH^-]$$

$$14 = pH + pOH$$

Practice: Fill in the table below with the missing information.

$[H^+]$	pH	$[OH^-]$	pOH	Acidic or Basic?
$2.5 \times 10^{-5} M$				
	7.92			
		$6.8 \times 10^{-4} M$		
			11.16	

pH of Strong Acids and Bases

- The pH of a solution of strong acid is simply calculated by taking the negative logarithm of concentration of a monoprotic strong acid.
- For example, the pH of 0.1 M HCl is _____
- Caution: If the acid concentration is less than 1.0×10^{-7} , the water becomes the important source of $[H^+]$ and the pH is 7.00. The pH of an acidic solution cannot be greater than _____ at 25°C!!!!

Practice:

With strong acids, the concentration of H^+ in solution the same as the concentration of the acid itself. (This is not the case with weak acids) Calculate the pH for each of the following concentrations of hydrochloric acid. Remember that if the concentration of $[\text{H}_3\text{O}^+]$ has two sig figs, the pH will have 2 decimal places.

Concentration of HCl	pH
0.50 M	
0.0075 M	
$3.4 \times 10^{-6} \text{ M}$	
$1.5 \times 10^{-8} \text{ M}$	

Calculate the concentration of nitric acid for each of the following solutions. Remember that if a pH has 2 decimal places, the concentration of $[\text{H}_3\text{O}^+]$ will have two sig figs.

Concentration of HNO_3	pH
	0.72
	2.04
	4.56

Fill in the missing information in the table below. Pay attention to the concentration of hydroxide in the last row of the table. $[\text{OH}^-]$ is not equal to 0.0050 M in 0.0050 M $\text{Ca}(\text{OH})_2$.

Solution	$[\text{OH}^-]$	pOH	pH
0.25 M NaOH			
0.0050 M KOH			
0.0050 M $\text{Ca}(\text{OH})_2$			

Fill in the missing information in the tables below. Remember that something extra should be done to convert from $[\text{OH}^-]$ into concentration of $\text{Sr}(\text{OH})_2$.

pH	pOH	Concentration of NaOH
10.23		
12.34		

pH	pOH	Concentration of $\text{Sr}(\text{OH})_2$
10.23		
12.34		

8.3 Weak Acid Base Equilibria

Acid Dissociation Reaction

Write a dissociation reaction for HCl, a STRONG acid, in water. Write the dissociation constant.

Write a dissociation reaction for HCN, a WEAK acid, in water. Write the dissociation constant.

Acid dissociation constant (K_a)

- Equilibrium expressions and conditions are used to calculate concentrations of H^+ , conjugate bases, and remaining acids
- The equilibrium expression for the dissociation of a weak acid is called the acid dissociation constant, or the K_a .

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} \quad \text{or} \quad K_a = \frac{[H^+][A^-]}{[HA]}$$

- Write the K_a expression for the weak acid, HCN.

Calculating pH of weak acids involves setting up an equilibrium. Weak acids have _____ K values, so an approximation for a small α can be used.

Always solve by:

- Writing the equation,
- Setting up the acid equilibrium expression (K_a),
- Defining initial concentrations, changes, and equilibrium concentrations in terms of α ,
- Substituting values and variables into the K_a expression and solving for α , and
- Using the α value to find the pH of the acid solution

Practice: Calculate the pH of a 0.100 M solution of acetic acid. The K_a of acetic acid is 1.8×10^{-5} .

Calculating K_a . Given the initial concentration and pH of a weak acid solution, the K_a can be calculated.

Practice: A 0.100 M weak acid (HA) solution has a pH of 4.25. Find K_a for the acid.

Determination of the Percent Dissociation (or Ionization) of a Weak Acid

- Percent Dissociation of a weak acid can be used to calculate pH, and vice versa
- % dissociation = $\frac{\text{amount dissociated (mol/L)}}{\text{initial concentration (mol/L)}} \times 100$

OR

$$\% \text{ dissociation} = \frac{\text{final } [\text{H}^+]}{\text{initial } [\text{HA}]} \times 100$$

- For a weak acid, percent dissociation (or ionization) _____ as the acid becomes more dilute (_____ molarity). Equilibrium shifts to the right for lower molarities.

Practice: Calculate the percent dissociation of the 0.100 M weak acid (HA) solution with pH of 4.25 (from the last problem.)

pKa and pKb

- Like pH and pOH, *pKa* and *pKb* express the values of K_a and K_b in a logarithmic function where $\text{pKa} = -\log K_a$ and $K_a = 10^{-\text{pKa}}$ and $\text{pKb} = -\log K_b$ and $K_b = 10^{-\text{pKb}}$
- Also, an acid and its conjugate base show the following relationship:
- $K_a \times K_b = K_w$ $\text{pKa} + \text{pKb} = 14$
- The above relationship is also true for a base and its conjugate acid

Practice: Find the pKa of acetic acid, whose $K_a = 1.8 \times 10^{-5}$.

What is the pKb of the conjugate base, $\text{C}_2\text{H}_3\text{O}_2^{-1}$?

What is the K_b of $\text{C}_2\text{H}_3\text{O}_2^{-1}$?

How are the K_a and the K_b related?

Calculating pH of polyprotic acids

- All polyprotic acids dissociate stepwise. Each dissociation has its own K_a value called K_{a1} , K_{a2} , K_{a3}
- As each H is removed, the remaining acid gets _____ and therefore has a _____ K_a . And as the negative charge on the acid increases it becomes more difficult to remove the positively charged proton.
- Because the _____ is much smaller than the _____ value, we treat ANY weak acid species as if it loses only 1 H^+ in significant quantities (only analyze the _____ H^+ removal)

Practice: Consider a 0.200 M H_2PO_4^- solution.

The $\text{p}K_a$ of H_2PO_4^- is 7.20 at 25°C.

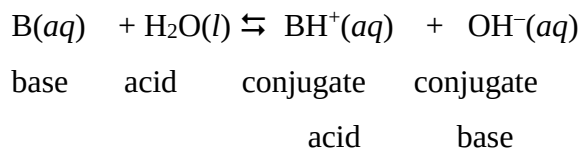
- List all ions present in the solution.
- Calculate the K_a of the acid.
- Write a dissociation reaction and equilibrium expression.
- Calculate the pH of the solution at 25°C.

Determination of the pH of a Mixture of Weak Acids

In a mixture of weak acids, only the acid with the _____ K_a value will contribute an appreciable $[\text{H}^+]$. Determine the pH based on this acid and _____ any others.

Analysis of Weak Bases

- Weak bases (bases without OH^-) react with water to produce a hydroxide ion. Common examples of weak bases are ammonia (NH_3), methylamine (CH_3NH_2), and ethylamine ($\text{C}_2\text{H}_5\text{NH}_2$).
- The lone pair on N forms a bond with a H^+ . Most weak bases involve N.
- The general form of the dissociation of a base in water:



Practice: Write the equation for the dissociation of NH_3 in water

Base dissociation constant (K_b)

- The general form for the base dissociation constant is:
- $K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$ and $\text{p}K_b = -\log K_b$

Practice: Write the K_b expression for the dissociation of NH_3 in water.

Analysis of pH of weak bases

- Determination of the pH of a weak base is very similar to the determination of the pH of a weak acid. Follow the same steps. Remember, however, that b is the $[\text{OH}^-]$ and taking the negative log of x will give you the pOH and not the pH!

Practice: Calculate the $[\text{OH}^-]$ and the pH for a 15.0 M NH_3 solution. The K_b for NH_3 is 1.8×10^{-5} .

Practice: Find the $[\text{OH}^-]$ and pH of a 0.100 M solution of pyridine, $\text{C}_5\text{H}_5\text{N}$. ($\text{p}K_b = 8.85$)

- Write a K_b expression, and find K_b from $\text{p}K_b$.
- Find the $[\text{OH}^-]$ and pH.

8.4 Acid Base Reactions and Buffers

Remember: Weak Acids and Bases have *significant* conjugates

- Because the conjugates of weak acids and bases are significant, their presence alone (without their partner) in water will cause the solution $\text{pH} \neq 7$
- Example: The salt NaCl_2O contains _____, the conjugate base of _____. The presence of this conjugate base would make the solution basic according to the following reaction:

Practice: Each of these salts contains a conjugate acid or base. Identify the significant conjugate acid or base, the partner, and hypothesize if a solution of salt would be acidic or basic or neutral.

Salt	Significant Species	Weak Partner	pH of salt solution?	Basic, Acidic or Neutral Salt?
NaCl_2O	Cl_2O^- (CB)	HCl_2O		
NH_4Cl				
NaNO_3				
$\text{KC}_2\text{H}_3\text{O}_2$				

Neutral Salts

- **Neutral Salts**- Salts that are formed from the cation of a strong base and the anion from a strong acid form neutral solutions when dissolved in water. Ex. NaCl , KNO_3
- The pH of these salts is always _____

Acidic Salts

- **Acid Salts**- Salts that are formed from the conjugate acid of a weak base and the anion from a strong acid form acidic solutions when dissolved in water.
- Ex. NH_4Cl (notice the N atom!) Write the reaction of the significant conjugate in water:
- The resulting solution will be _____ because NH_4^+ donates a proton from water

Basic Salts

- **Basic Salts** - Salts that are formed from the conjugate base of a weak acid and the neutral cation form basic solutions when dissolved in water.

Ex. NaNO_2 Write the reaction of the significant conjugate in water:

- The resulting solution will be _____ because NO_2^- accepts a proton from water

General Order of pH

High pH
↑
Low pH

Strong Bases (SB)
Bases (WB)
Salts (BS)
Neutral Salts (NS)
Salts (AS)
Acids (WA)
Strong Acids (SA)

Notice that in general, salts are weaker than weak species, which are weaker than strong species!

Weak
Basic

Acidic
Weak

Classify each type of species according to the system above. Arrange the following 0.1 M solutions in order of decreasing pH by numbering 1-7. Write a dissociation reaction for each that shows the affect on pH.

<u>Substance</u>	<u>Species type?</u>	<u>Rank</u>	<u>Dissociation in water</u>
NaOH	SA		$\text{NaOH} + \text{H}_2\text{O} \longrightarrow$
NaCN			
KCl			
CH_3NH_2			
H_3PO_4			
NH_4NO_3			
HBr			

Determination of the pH of Salts

- When determining the exact pH of salt solutions, we can find the K value of the significant conjugate from the K value of the weak partner. They are related with the equation:

$$K_a \times K_b = K_w \quad \text{Also,} \quad pK_a + pK_b = 14$$

$$\text{where } pK_a = -\log K_a$$

$$\text{and } pK_b = -\log K_b$$

- K_a of a significant conjugate acid can be found from the K_b of the weak base partner.
- K_b of a significant conjugate base can be found from the K_a of the weak acid partner.

Practice: If NH_3 has a K_b value of 1.8×10^{-5} , what is the K_a needed to analyze the salt NH_4Cl ?

Practice: Calculate the pH of a 0.15 M solution of sodium acetate. (First, determine what $\text{NaC}_2\text{H}_3\text{O}_2$ is!)

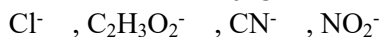
The K_a of acetic acid is 1.8×10^{-5} .

Practice: Find the pH of a 0.100 M $\text{CH}_3\text{NH}_3\text{Cl}$ solution. The pK_b for CH_3NH_2 is 3.35.

Practice:

$\text{HC}_2\text{H}_3\text{O}_2$	1.8×10^{-5}
HCN	6.2×10^{-10}
HNO_2	4.0×10^{-4}

Given the K_a values for the acids above, rank these conjugate bases from weakest to strongest.



13

8.6 Molecular Structure of Acids and Bases

Hydrohalic acids- Relative Acid Strength

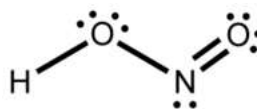
- Hydrohalic acids- H is attached to a halogen (HCl, HI, HBr, HF)
- HF is the only weak hydrohalic acid. Although the H-F bond is very polar, the bond is so strong (due to the small F atom) that the acid does not completely dissociate.
- The others (HCl, HBr, HI) have larger anions, and the larger the distance, the weaker the attraction between H^+ and X^- weaker (Coulomb's Law)
- Of the hydrohalic acids, which is the strongest?

Relative Acid Strength

- **Oxyacids**- most acids are oxyacids

Examples of weak oxyacids:

- H_3PO_4 (phosphoric), HNO_2 (nitrous), HOCl (hypochlorous)



- acidic proton is attached to O

Oxyacid Relative Acid Strength- multiple O's

- In general, the more electron dense an anion is, the looser the H-A bond is, and the stronger the acid is.
- Within a series, acid strength increases with increasing numbers of oxygen atoms. For example:
 $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$
- and $\text{H}_2\text{SO}_4 > \text{H}_2\text{SO}_3$

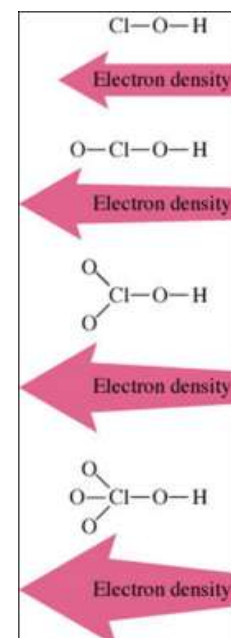
(Electronegative O draws electrons away from O-H bond)

- Match the acid with the K_a

- HIO_3	2×10^{-11}
- HIO	1.6×10^{-1}
- HIO_2	3.2×10^{-5}

Oxyacid Relative Acid Strength- varying central atom

- Acid strength increases with increasing electronegativity of oxyacids.
the acids with decreasing strength:



Buffer Solutions and Common ion effect

- The addition of an ion already present (common) in a system cause
- Example: The addition of a common ion to a solution of a weak acid makes the solution less acidic. Write the dissociation reaction for acetic acid:
- Using the LeChatlier Principle, determine how the addition of $\text{NaC}_2\text{H}_3\text{O}_2$ causes the equilibrium to shift. Will the pH increase or decrease?

Buffer Solution

- A buffer solution is made by combining a _____ acid or base with a _____ containing its _____.
- When preparing a buffer, choose an acid whose _____ is close to the desired pH of the buffer solution.
- Alternatively, the _____ of the base should be close to the pOH of the buffer solution.
- These solutions are called buffers because they _____ a change in pH when acid or base is added.

Which combinations of compounds can make a buffer solution?

- HCHO_2 and NaCHO_2
- HCl and NaCl
- CH_3NH_2 and $\text{CH}_3\text{NH}_3\text{Cl}$
- NH_3 and NaOH

Weak Acids	pKa	Weak Bases	pKb	Salts
Chlorous acid, HClO_2	1.92	Methylamine, CH_3NH_2	3.36	NaClO_2
Nitrous acid, HNO_2	3.40	Ammonia, NH_3	4.74	NaHCOO
Formic acid, HCOOH	3.74	Pyridine, $\text{C}_5\text{H}_5\text{N}$	8.77	KNO_2
Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$	4.19			$\text{KC}_6\text{H}_5\text{COO}$
Acetic acid, CH_3COOH	4.74			NaCH_3COO
Hypochlorous acid, HClO	7.46			NaClO
Hydrocyanic acid, HCN	9.21			KCN
				$\text{CH}_3\text{NH}_3\text{NO}_3$
				NH_4Cl
				$\text{C}_2\text{H}_5\text{NHCl}$

Practice: You have been asked to prepare 500.0 mL of **buffer solution** at pH 4.00. A student has suggested using ammonia, NH_3 , and ammonium chloride, NH_4Cl , to prepare the buffer. Explain why this is NOT an appropriate choice to prepare the buffer.

From the table above, what would be the *best* combination to prepare the buffer?

Buffer solutions resist pH change by reaction with strong acid or base

Example: Consider a mixture of $\text{HC}_2\text{H}_3\text{O}_2$ and $\text{NaC}_2\text{H}_3\text{O}_2$

Write an equation for the addition of strong acid... hint, which component will it react with? (*Easiest way to show a strong acid is to use H^+*):

Write an equation for the addition of strong base ... hint, which component will it react with? (*Easiest way to show a strong acid is to use OH^-*):

Practice: Consider an $\text{NH}_3/\text{NH}_4^+$ buffer system

Write an equation for the addition of strong acid to the system:

Write an equation for the addition of strong acid to the system:

Buffer capacity

- The amount of acid or base that can be absorbed by a buffer system without a significant change in pH. In order to have a _____ buffer capacity, a solution should have _____ concentrations of both buffer components.

Which has a larger capacity and why?

0.10 M HF/0.10 M NaF, pK_a 3.14

0.01 M HClO_2 / 0.01 M KClO_2 pK_a 1.95

Calculating the pH of a Buffer System

One way to calculate the pH of a buffer system is with the **Henderson-Hasselbach** equation.

$$\text{pH} = \text{pK}_a + \log \frac{[\text{base}]}{[\text{acid}]}$$

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Creating Buffers

- For a particular buffering system _____ solutions that have the same ratio of $[\text{A}^-]/[\text{HA}]$ have the same _____.
- Optimum buffering occurs when _____ and the _____ of the weak acid used should be as close as possible to the desired pH of the buffer system.

Formic acid (HCHO_2) has a $\text{pK}_a = 3.74$. Calculate the pH of a buffer that is 0.25 M HCHO_2 and 0.25 M NaCHO_2 .

- Notice that whenever _____, $\text{pH} = \text{_____}$!

Hints for Solving Buffer Problems:

1. Determine major species involved initially.
2. If chemical reaction occurs(Is there a strong acid or strong base?), write equation and solve stoichiometry in moles, then change to molarity.
3. Write equilibrium equation.
4. Set up equilibrium expression (K_a or K_b).
5. Solve.
6. Check logic of answer.

Practice: A solution is 0.120 *M* in acetic acid and 0.0900 *M* in sodium acetate. Calculate the $[H^+]$ at equilibrium. The K_a of acetic acid is 1.8×10^{-5} . Solve using a RICE table.

Now solve using H-H equation:Ex. A solution is 0.120 *M* in acetic acid and 0.0900 *M* in sodium acetate. Calculate the $[H^+]$ at equilibrium. The K_a of acetic acid is 1.8×10^{-5} .

Practice: Calculate the pH of a solution that is 0.250 *M* in HCN and 0.170 *M* in KCN. For HCN, $K_a = 4.9 \times 10^{-10}$. Use either the equilibrium approach and the Henderson–Hasselbach approach.

Follow up: A strong base is added to the buffer solution. Write an equation showing the addition of the base to the buffer. Predict whether the pH will increase, decrease or stay the same.

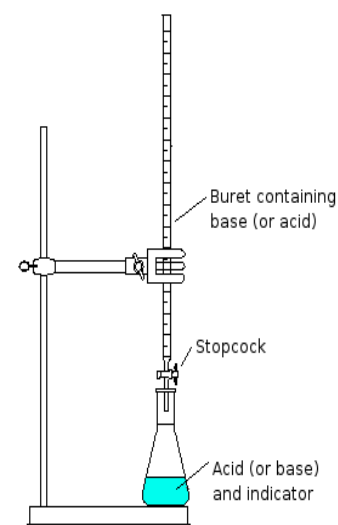
Neutralization Rxns

Predict the products, determine if pH <7, >7, =7 at completion (careful! Notice the arrow! Acid + Base has a LARGE K value!):

- $\text{NaOH} + \text{HCl} \rightarrow$
- $\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow$
- $\text{NH}_3 + \text{HI} \rightarrow$
- $\text{KOH} + \text{HC}_2\text{H}_3\text{O}_2 \rightarrow$
- $\text{NaOH} + \text{H}_3\text{PO}_4 \rightarrow$

Titrant and Analyte

- titrant-solution of known concentration (in buret)
- The titrant is added to a solution of unknown concentration (analyte) until the substance being analyzed is just consumed (stoichiometric point or equivalence point).
- The equivalence point is where the moles of titrant is just enough to completely neutralize the moles of analyte
- In a titration, the data collected can include initial and final volume of the titrant, molarity of the titrant, and the volume of the analyte.



Practice: The titration of 25.00 mL of acid HA was performed with a 0.550 M solution of NaOH. The buret initially read 1.25 mL and showed a final volume of 24.76 mL.

- What is the titrant and what is the analyte?
- Calculate the concentration of the acid.

The progress of a Neutralization Reaction is commonly charted on a pH or Titration Curve

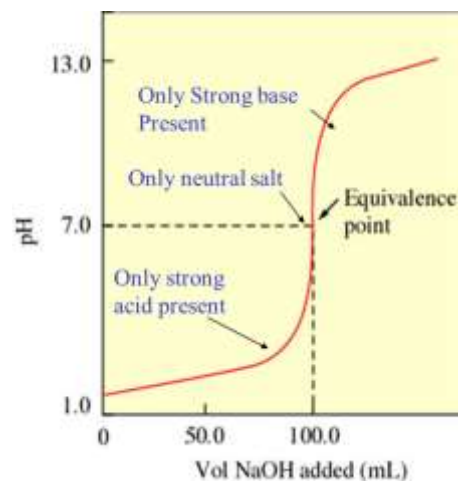
- pH or titration curve -plot of pH as a function of the amount of titrant added.
- The x axis represents _____
- The y axis represents _____
- The equivalence point is the _____ of the _____ section of the curve

Simplest Curve: Strong acid-strong base

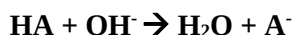
- Simple reaction $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
- The pH is easy to calculate because all reactions go to completion.
- At the equivalence point, the solution is _____, pH is _____.

The pH Curve for the Titration of 50.0 mL of HNO_3 with 0.100 M NaOH

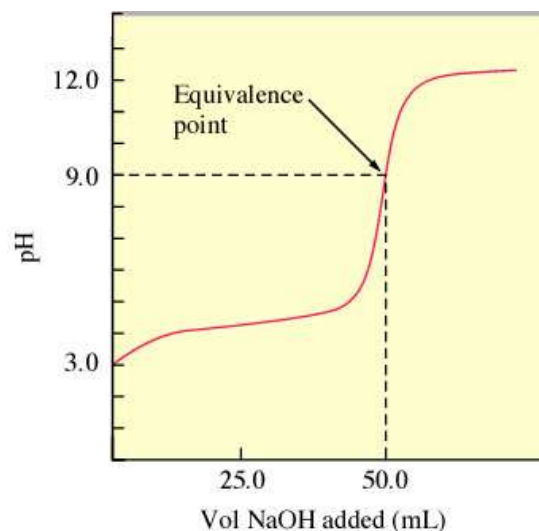
- Write a reaction for the titration:
- What is the pH at the equivalence point? Why?
- Using the equivalence point data, calculate the molarity of the acid.



Weak acid - strong base

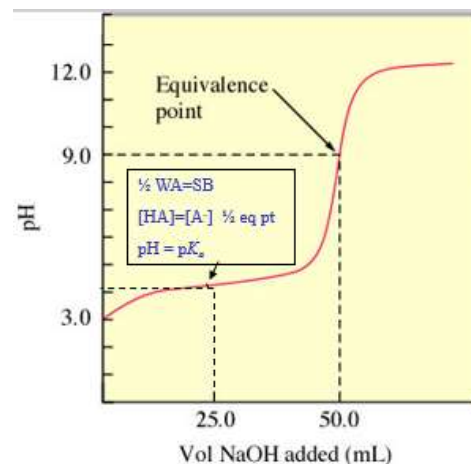


- When titrating a weak acid with a strong base, there are several important points:
- **Initial pH- What species are present at this point? How would we find the pH?**
- Because the solution has only weak acid (dominant species), pH can be determined using the K_a , _____.
- **Equivalence Point- What species are present at this point?**
At the equivalence, the conjugate base of the weak acid is dominant species. That's why the pH is _____. The exact pH can be calculated like a _____
- Easiest way to tell this is a weak acid titration? $\sim \text{pH}$ at Equivalence point is _____!!!



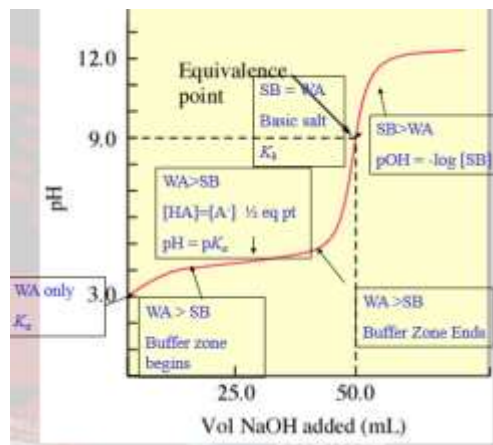
When titrating a weak acid with a strong base, there are several important points:

- **Half Equivalence Point: What is meant by 'half'?**
- At halfway to the equivalence point:
 - 50% of the acid remains and 50% of the conjugate base is present
 - Therefore, _____ at half equivalence point!
 - pK_a can be determined from a titration curve in this way, and vice versa
 - What is the pK_a of this weak acid? _____



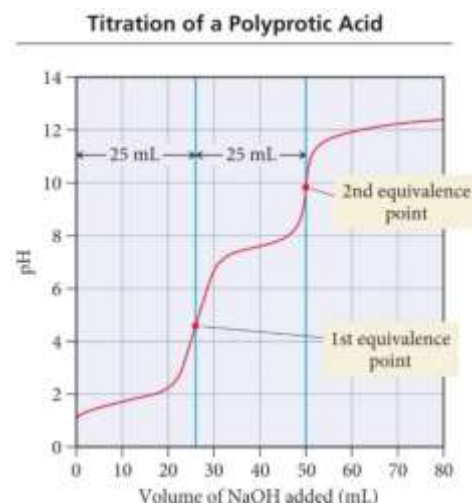
Other points of interest:

- **Buffer Zone:** between about 12 mL to 37 mL on this graph- surrounds the $\frac{1}{2}$ equivalence point- both HA and A^- are present in large enough quantities that buffering occurs- pH changes very little
- **Beyond the Equivalence Point:** all acid is neutralized, only strong base remains, pH is found by excess moles of base present



Titration of a Polyprotic Acid

- If $K_{a1} \gg K_{a2}$, there will be two equivalence points in the titration.
 - Graph has a 'zigzag' pattern
 - The closer the K_a 's are to each other, the less distinguishable the equivalence points are.
 - Notice that it takes equal amounts of moles to reach each equivalence point



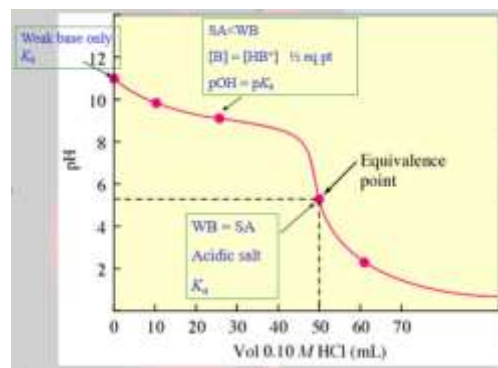
Weak base - Strong acid Titration



- Before the equivalence point, a weak base equilibria exists.
- At $\frac{1}{2}$ equivalence point _____
- At equivalence point, the conjugate acid causes the solution to have a _____

Circle the buffer zone

Easiest way to tell this is a weak acid titration?



Indicators

- Indicators are usually weak acids, _____. They have one color in their acidic (_____) form and another color in their basic (_____) form.
- $H^+ + In^- \rightleftharpoons HIn$ ~OR~ $HIn + OH^- \rightleftharpoons H_2O + In^-$
- A very common indicator, phenolphthalein, is colorless in its HIn form and pink in its In^- form. It changes color in the range of pH 8-10.

Choosing an indicator

- The useful range of an indicator is usually the indicator $pK_a \pm 1$ (the buffering region).
- When choosing an indicator, determine the pH at the equivalence point of the titration and then choose an indicator with a pK_a close to that.

(c) The pH at the equivalence point of the titration is measured to be 2.54. Which of the following indicators would be the best choice for determining the end point of the titration? Justify your answer.

Indicator	pKa
Phenolphthalein	9.3
Bromothymol blue	7.0
Methyl red	5.0
Thymol blue	2.0
Methyl violet	0.80

Acid-Base Equilibria

Identifying Acids, Bases, and Conjugates

1. List the correct Conjugate acids and conjugate bases for the acids and bases given below.

Acid	Conjugate base		Base	Conjugate Acid
a. HCl			a. NH ₃	
HNO ₃			HSO ₄ ⁻	
b. H ₂ O			b. OH ⁻	
c. H ₂ SO ₄			c. HCO ₃ ⁻	
d. H ₂ S			d. NO ₃ ⁻	

2. From each of the following reactions, select the substance which is the conjugate acid and the substance which is the conjugate base. Put an A above the acid and a B above the base.

Equation	Conj. Base	Conj. Acid
a) $\text{NH}_3 + \text{H}_3\text{O}^+ \rightarrow \text{NH}_4^+ + \text{H}_2\text{O}$	_____	_____
b) $\text{CH}_3\text{OH} + \text{NH}_2^- \rightarrow \text{CH}_3\text{O}^- + \text{NH}_3$	_____	_____
c) $\text{OH}^- + \text{H}_3\text{O}^+ \rightarrow \text{H}_2\text{O} + \text{H}_2\text{O}$	_____	_____
d) $\text{NH}_2^- + \text{H}_2\text{O} \rightarrow \text{NH}_3 + \text{OH}^-$	_____	_____
e) $\text{H}_2\text{O} + \text{HClO}_4 \rightarrow \text{H}_3\text{O}^+ + \text{ClO}_4^-$	_____	_____

3. The measure of the hydrogen ion concentration in a solution is referred to as _____.

4. Circle whether the following is acidic, basic, or neutral.

a) pH= 4.0	acidic	basic	neutral
b) pH= 12.0	acidic	basic	neutral
c) pOH= 2.0	acidic	basic	neutral
d) $[\text{OH}^-] = 1.1 \times 10^{-3}$	acidic	basic	neutral
e) $[\text{H}^+] = 1.0 \times 10^{-9}$	acidic	basic	neutral

5. Complete the following table:

pH	pOH	$[\text{H}^+]$	$[\text{OH}^-]$
2.50			
	8.20		
		2.6×10^{-1}	
			6.2×10^{-10}

pH PRACTICE

8.8×10^{-1}	2×10^{-4}	1×10^{-7}	5×10^{-9}	3.6×10^{-11}
$\boxed{\text{H}^+}$	$\boxed{\text{H}^+}$	$\boxed{\text{H}^+}$	$\boxed{\text{H}^+}$	$\boxed{\text{H}^+}$
				$\boxed{\text{OH}^-}$
$\boxed{\text{OH}^-}$	$\boxed{\text{OH}^-}$	$\boxed{\text{OH}^-}$	$\boxed{\text{OH}^-}$	$\boxed{\text{OH}^-}$

- Write the pH of each solution above the $[\text{H}^+]$'s. $\text{pH} = -\log[\text{H}^+]$
- Label the “Z” diagram as “Acidic”, “Basic” and “Neutral.”
- Knowing that the $[\text{H}^+] \times [\text{OH}^-]$ always equals 1×10^{-14} , fill in the $[\text{OH}^-]$ for each of the five solutions in the “Z” Diagram. (1×10^{-14} is called the Dissociation Constant for water, K_w) ($1 \times 10^{-14} = 10 \times 10^{-15}$)
- Write the “pOH” of each solution below the $[\text{OH}^-]$'s.
- $\text{pH} + \text{pOH}$ always equals_____.
- A solution of acid has $[\text{H}^+] = 3.0 \times 10^{-3}\text{M}$
 - Calculate the $[\text{OH}^-]$ _____
 - Calculate the pH_____ the pOH _____
- A solution of base has an $[\text{OH}^-] = 4.25 \times 10^{-5}\text{M}$
 - Calculate the $[\text{H}^+]$ _____
 - Calculate the pH_____ the pOH _____
- Calculate the pH's of the following solutions:

$2.53 \times 10^{-2}\text{M HCl}$ $\text{pH} =$

$2.53 \times 10^{-4}\text{M HCl}$ $\text{pH} =$

$2.53 \times 10^{-5}\text{M HCl}$ $\text{pH} =$

A pH with 3 significant figures is written with_____numbers after the decimal place.

pH of Strong and Weak Species

1. **strong acid solution** – determine $[H^+]$, calculate pH
Calculate the pH of 0.00125M HNO_3
2. **strong base solution** – determine $[OH^-]$, calculate pOH, calculate pH
Calculate the pH of 0.00125M KOH
3. **weak acid solution** – determine $[H^+]$ using ICE box, calculate pH
Calculate the pH of 0.00125M $HOCl$ $K_a = 3.5 \times 10^{-8}$
4. **weak base solution** – determine $[OH^-]$ using ICE box, calculate pOH, calculate pH
Calculate the pH of 0.00125M NH_3 $K_b = 1.8 \times 10^{-5}$
5. **salt of a weak acid** – write hydrolysis, calc K_b , determine $[OH^-]$ using ICE box, calc pOH, calc pH
Calculate the pH of 0.00125M $NaOCl$

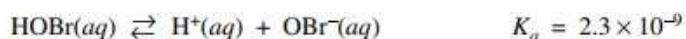
6. **salt of a weak base** – write hydrolysis, calc K_a , determine $[H^+]$ using ICE box, calc pH
Calculate the pH of 0.00125M NH_4Cl
7. **diprotic acid solution** – assume all $[H^+]$ from first ionization, determine $[H^+]$ using ICE box, calculate pH
Calculate the pH of 0.00125M H_2CO_3 $K_{a1} = 4.2 \times 10^{-7}$ $K_{a2} = 4.8 \times 10^{-11}$
8. **mixture of acid and base** – calculate moles of H^+ and OH^- , determine moles of excess H^+ or OH^- , determine total volume, calculate $[H^+]$ or $[OH^-]$, calculate pH
Calculate the pH of 20.0 mL of 0.00125M HNO_3 + 30.0 mL of 0.00125M KOH

Answers:

1	2	3	4	5	6	7	8
pH = 2.903	pH = 11.097	pH = 5.18	pH = 10.15	pH = 9.28	pH = 6.08	pH = 4.64	pH = 10.398

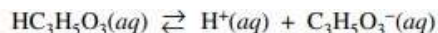
Unit 8.1 – Practice AP Chem Probs

2002 #1- you can do parts a, b, e for this unit, parts c and d after Unit 8.2



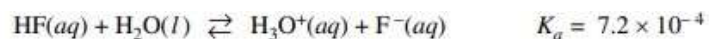
1. Hypobromous acid, HOBr, is a weak acid that dissociates in water, as represented by the equation above.
- Calculate the value of $[\text{H}^+]$ in an HOBr solution that has a pH of 4.95.
 - Write the equilibrium constant expression for the ionization of HOBr in water, then calculate the concentration of HOBr(aq) in an HOBr solution that has $[\text{H}^+]$ equal to $1.8 \times 10^{-5} M$.
 - A solution of $\text{Ba}(\text{OH})_2$ is titrated into a solution of HOBr.
 - Calculate the volume of $0.115 M \text{Ba}(\text{OH})_2(aq)$ needed to reach the equivalence point when titrated into a 65.0 mL sample of $0.146 M \text{HOBr}(aq)$.
 - Indicate whether the pH at the equivalence point is less than 7, equal to 7, or greater than 7. Explain.
 - Calculate the number of moles of $\text{NaOBr}(s)$ that would have to be added to 125 mL of $0.160 M \text{HOBr}$ to produce a buffer solution with $[\text{H}^+] = 5.00 \times 10^{-9} M$. Assume that volume change is negligible.
 - HOBr is a weaker acid than HBrO_3 . Account for this fact in terms of molecular structure.

2002 Form B #1- you can do parts a and b for this unit, c and d after unit 8.2



1. Lactic acid, $\text{HC}_3\text{H}_5\text{O}_3$, is a monoprotic acid that dissociates in aqueous solution, as represented by the equation above. Lactic acid is 1.66 percent dissociated in $0.50 M \text{HC}_3\text{H}_5\text{O}_3(aq)$ at 298 K. For parts (a) through (d) below, assume the temperature remains at 298 K.
- Write the expression for the acid-dissociation constant, K_a , for lactic acid and calculate its value.
 - Calculate the pH of $0.50 M \text{HC}_3\text{H}_5\text{O}_3$.
 - Calculate the pH of a solution formed by dissolving 0.045 mole of solid sodium lactate, $\text{NaC}_3\text{H}_5\text{O}_3$, in 250. mL of $0.50 M \text{HC}_3\text{H}_5\text{O}_3$. Assume that volume change is negligible.
 - A 100. mL sample of $0.10 M \text{HCl}$ is added to 100. mL of $0.50 M \text{HC}_3\text{H}_5\text{O}_3$. Calculate the molar concentration of lactate ion, $\text{C}_3\text{H}_5\text{O}_3^-$, in the resulting solution.

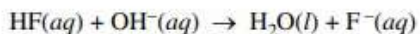
2007 #1- you can do parts a, b, now, c,d,e after Unit 8.2



1. Hydrofluoric acid, $\text{HF}(aq)$, dissociates in water as represented by the equation above.

- (a) Write the equilibrium-constant expression for the dissociation of $\text{HF}(aq)$ in water.
- (b) Calculate the molar concentration of H_3O^+ in a 0.40 M $\text{HF}(aq)$ solution.

$\text{HF}(aq)$ reacts with $\text{NaOH}(aq)$ according to the reaction represented below.



A volume of 15 mL of 0.40 M $\text{NaOH}(aq)$ is added to 25 mL of 0.40 M $\text{HF}(aq)$ solution. Assume that volumes are additive.

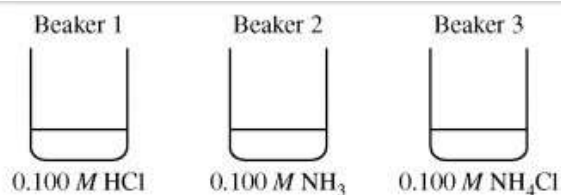
- (c) Calculate the number of moles of $\text{HF}(aq)$ remaining in the solution.
- (d) Calculate the molar concentration of $\text{F}^-(aq)$ in the solution.
- (e) Calculate the pH of the solution.

1. Answer the following questions that relate to the chemistry of halogen oxoacids.

(a) Use the information in the table below to answer part (a)(i).

Acid	K_a at 298 K
HOCl	2.9×10^{-8}
HOBr	2.4×10^{-9}

- (i) Which of the two acids is stronger, HOCl or HOBr? Justify your answer in terms of K_a .
 - (ii) Draw a complete Lewis electron-dot diagram for the acid that you identified in part (a)(i).
 - (iii) Hypoiodous acid has the formula HOI. Predict whether HOI is a stronger acid or a weaker acid than the acid that you identified in part (a)(i). Justify your prediction in terms of chemical bonding.
- (b) Write the equation for the reaction that occurs between hypochlorous acid and water.
- (c) A 1.2 M NaOCl solution is prepared by dissolving solid NaOCl in distilled water at 298 K. The hydrolysis reaction $\text{OCl}^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HOCl}(aq) + \text{OH}^-(aq)$ occurs.
- (i) Write the equilibrium-constant expression for the hydrolysis reaction that occurs between $\text{OCl}^-(aq)$ and $\text{H}_2\text{O}(l)$.
 - (ii) Calculate the value of the equilibrium constant at 298 K for the hydrolysis reaction.
 - (iii) Calculate the value of $[\text{OH}^-]$ in the 1.2 M NaOCl solution at 298 K.



1. Each of three beakers contains 25.0 mL of a 0.100 M solution of HCl, NH₃, or NH₄Cl, as shown above. Each solution is at 25°C.
- (a) Determine the pH of the solution in beaker 1. Justify your answer.
 - (b) In beaker 2, the reaction $\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$ occurs. The value of K_b for $\text{NH}_3(aq)$ is 1.8×10^{-5} at 25°C.
 - (i) Write the K_b expression for the reaction of $\text{NH}_3(aq)$ with $\text{H}_2\text{O}(l)$.
 - (ii) Calculate the $[\text{OH}^-]$ in the solution in beaker 2.
 - (c) In beaker 3, the reaction $\text{NH}_4^+(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_3(aq) + \text{H}_3\text{O}^+(aq)$ occurs.
 - (i) Calculate the value of K_a for $\text{NH}_4^+(aq)$ at 25°C.